

UNIT 4

1. Let  $f(x) = \frac{2x+1}{3-x}$

- a) Prove this is one to one by using the definition of one to one-ness.
- b) Find  $f^{-1}(x)$
- c) Sketch  $f(x)$

2. Let  $f(x) = \sqrt{x^2 - 1}$  and  $g(x) = \frac{1}{2x-6}$

- a) Find the domain of each function above.
- b) Find the domain of each of the following compositions:

i)  $(f+g)(x)$       ii)  $\left(\frac{g}{f}\right)(x)$       iii)  $(g \circ f)(x)$

3. Sketch  $g(x) = \lfloor -|x| \rfloor$

4. If the point  $(2, -3)$  is on the graph of a function  $f(x)$ , what point(s) must lie on the graph of:

- a)  $f(2x)$
- b)  $f(3x - \frac{1}{2})$
- c)  $f(-x)$
- d)  $|f(x)|$
- e)  $f(|x|)$
- f)  $f^{-1}(x)$

5. If  $f(x) = \{(0,1), (1,-2), (2,3), (3,-1)\}$  and  $g(x) = \{(1,2), (-1,1), (3,0)\}$ , find the domain of  $f \circ g$  and  $g \circ f$ .

6. Given that  $f(x)$  is even and  $g(x)$  is odd, determine, with proof, if the following are odd, even or neither:

- a)  $h(x) = g(x) + x^3$
- b)  $k(x) = g(g(x))$
- c)  $p(x) = g(x) + 5$
- d)  $q(x) = f(g(x))$

## UNIT 5

0. What does the Remainder Theorem state? What does the Factor Theorem state? What does the Rational Roots Theorem state? What does the complex conjugate theorem state?

1.  $P(x) = x^4 - 5x^3 + 8x^2 - 5x + 1$ .

a. Solve:  $P(x) \geq 0$ .

b. Give the monomial which determines the global behavior.

c. Give the functions that describe local behavior near each integer x-intercept.

d. Graph  $P(x)$ .

2.

a) Build a polynomial in standard form and of lowest degree such that  $P(2i)=0$ ,  $P(3)=0$ .

b) Build a polynomial with real coefficients in standard form and of lowest degree such that  $P(2i)=0$ ,  $P(3)=0$  and  $P(1)=20$ .

3.  $f(x) = \frac{x^4 - 3x^3 + 3x^2 - x}{x^2 - 3x}$

a. Find the x intercepts.

b. Find the domain restrictions and identify them as VA or holes.

c. Solve  $f(x) \geq 0$ .

4. Write a polynomial function for which all of the following holds:

- End behavior: As  $x \rightarrow \infty, y \rightarrow \infty$ . Also  $x \rightarrow -\infty, y \rightarrow -\infty$
- Local behavior: An x intercept sits at  $x=1$ . Near  $x=1$ , function behaves cubically.
- Local behavior: An x intercept sits at  $x=3$ . Near  $x=3$ , function behaves quadratically.

Now give 2 other equations which would meet all these requirements as well but which don't introduce any other x intercepts.

5. If  $f(x) = x^3 + 2x^2 - 5x - 6$ , show that  $x+3$  is a factor using two different methods?

6. Let  $g(x) = x^5 - 6x^3 - 8x^2 - 3x$ ; if  $x=-1$  is a root of multiplicity greater than 1, find the following:

a) all the zeros

b) The monomial that  $g(x)$  behaves like globally, and show the end behavior using limit notation

c) What function does  $g(x)$  behave like near each x-intercept?

d) Perform a sign analysis for  $g(x)$

e) Graph  $g(x)$

7. Writing equations of functions.

a) Write a polynomial function

- whose coefficients are rational
- which is in standard form
- which is of lowest degree
- whose zeros include  $x=0$ ,  $x=1-\sqrt{3}$
- which runs through the point (1,5)

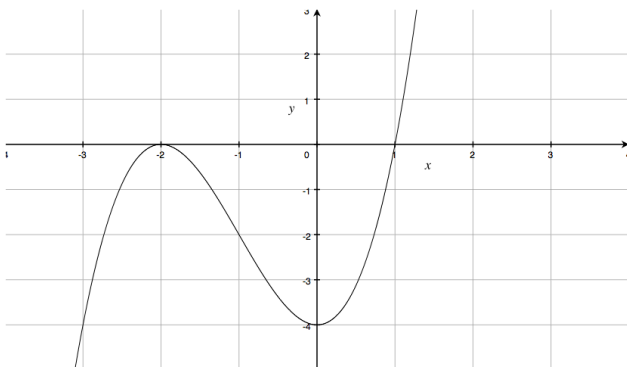
b) Write a rational function equation

- with a hole at the point (4,-2)
- with a vertical asymptote of  $x=2$
- with an x intercept of  $x=1$
- with a horizontal asymptote. (determine what it must be)

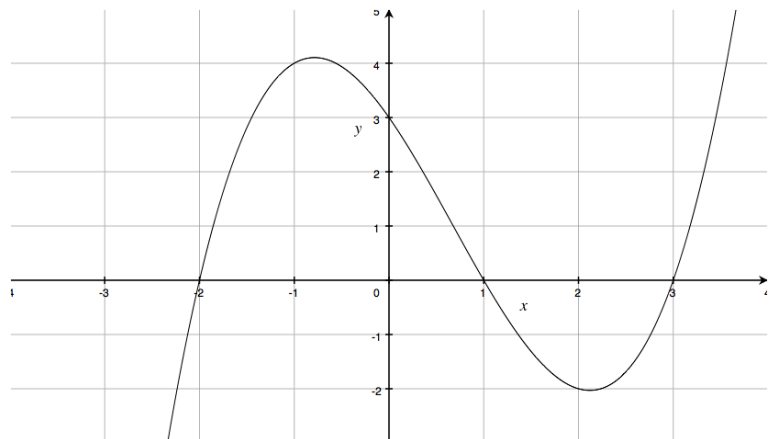
c) Write a polynomial function which equals 7 at  $x=2$ , 3 and 4 such that  $P(1)=14$

8.  $f(x)$  and  $g(x)$  are pictured below; graph  $p(x) = \frac{f(x)}{g(x)}$

**f(x)**



**g(x)**



9. Given that  $2 - i\sqrt{3}$  is a zero of the polynomial  $P(x) = x^4 - 6x^3 + 16x^2 - 18x + 7$ , find all the remaining zeros of  $P(x)$ .